

Md Rifat Arefin

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Intro

I am a Ph.D. candidate at Université de Montréal and Mila, supervised by Irina Rish. I study how architecture and optimization shape the representations and training dynamics of large language models, combining theory, mainly tensor decomposition and random matrix theory, with controlled experiments at scale. Recent work covers optimization dynamics in LLM pretraining with Muon, the expressivity of gated MLP blocks such as SwiGLU, KV-cache compression, and representation collapse and its effect on multi-step reasoning.

Work Experience

Research Scientist Intern:

Meta; Bellevue, WA, USA

June 2026 to November 2026

Research Scientist Intern:

Meta; California, USA

August 2025 to January 2026

Studied LLM pretraining dynamics with the Muon optimizer. Showed that temporary weight decay can hurt loss when it ends yet improve it at the final budget, and that a schedule selected at 162M parameters transfers without retuning to 1.4B (preprint, 2026).

Research Intern:

Samsung; Montreal, Canada

February 2025 to July 2025

Worked on improving reasoning in large language models through architectural modifications and training strategies for long-context molecule generation.

Applied Scientist Intern:

Amazon; Seattle, USA

October 2024 to January 2025

Worked on diffusion-based generative models for controllable image generation, improving conditioning techniques to enhance fidelity, diversity, and user-driven customization.

Visiting Researcher (part-time):

ServiceNow; Montreal, Canada

May 2024 to September 2024

Conducted research on improving reasoning in large language models, focusing on architectural modifications and training techniques for structured problem solving.

Research Intern (part-time):

Recursion Pharmaceuticals; Montreal, Canada

June 2023 to December 2023

Learned representations from images of human cells perturbed by CRISPR/Cas9 gene knockouts, compounds, and other reagents, to predict relationships across biology and chemistry.

Visiting Researcher:

Eberhard Karls University of Tübingen; Tübingen, Germany

January 2021 to June 2021

Worked on speech separation conditioned on speaker style, published at Interspeech 2021.

Deep Learning R&D Engineer:

UpStride; Paris, France

February 2020 to November 2020

Explored alternative algebras such as complex numbers and quaternions for deep learning architectures, aiming to improve the efficiency and expressiveness of neural networks.

Research Intern:

Mila – Quebec AI Institute; Montreal, Canada

August 2018 to February 2019

Research under the supervision of Yoshua Bengio on multi-image satellite super-resolution. Team “rarefin” placed 1st runner-up in the European Space Agency’s PROBA-V Super-Resolution competition (<https://kelvins.esa.int/proba-v-super-resolution/leaderboard/results>).

Intern:

Orion Informatics Limited; Dhaka, Bangladesh

January 2016 to June 2016

Six-month internship for my Bachelor’s degree, working on software and web application development.

Education

Mila – Quebec AI Institute & Université de Montréal

Expected graduation: August 2027

Ph.D. Candidate, Computer Science

Sep 2021 – Aug 2027

Supervisor: Irina Rish. CGPA 4.3 / 4.3.

Kyung Hee University

Department of Computer Science & Engineering

M.Sc. in Computer Science & Engineering

Mar 2017 – Feb 2019

CGPA 4.3 / 4.3.

University of Dhaka

Institute of Information Technology

B.Sc. in Software Engineering

Jan 2013 – Feb 2017

CGPA 3.89 / 4.

Publications

Conference papers.....

- [1] **Md Rifat Arefin**, Farzaneh Heidari, Irina Rish, Guillaume Rabusseau. *CP-MLPs: A Tensor-Rank Theory of Tied and Untied MLP Blocks*. Neural Information Processing Systems (**NeurIPS**), 2026.
- [2] Oscar Skea, **Md Rifat Arefin**, Dan Zhao, Niket Nikul Patel, Jalal Naghiyev, Yann LeCun, Ravid Shwartz-Ziv. *Layer by Layer: Uncovering Hidden Representations in Language Models*. International Conference on Machine Learning (**ICML**), 2025. **Spotlight**.
- [3] **Md Rifat Arefin**, Gopesh Subbaraj, Nicolas Gontier, Yann LeCun, Irina Rish, Ravid Shwartz-Ziv, Christopher Pal. *Seq-VCR: Preventing Collapse in Intermediate Transformer Representations for Enhanced Reasoning*. International Conference on Learning Representations (**ICLR**), 2025.
- [4] **Md Rifat Arefin**, Yan Zhang, Aristide Baratin, Francesco Locatello, Irina Rish, Dianbo Liu, Kenji Kawaguchi. *Unsupervised Concept Discovery Mitigates Spurious Correlations*. International Conference on Machine Learning (**ICML**), 2024.

[5] Oleksiy Ostapenko, Timothée Lesort, Pau Rodríguez, **Md Rifat Arefin**, Arthur Douillard, Irina Rish, Laurent Charlin. *Continual Learning with Foundation Models: An Empirical Study of Latent Replay*. Conference on Lifelong Learning Agents (**CoLLAs**), 2022.

[6] Shell Xu Hu, **Md Rifat Arefin**, Viet-Nhat Nguyen, Alish Dipani, Xaq Pitkow, Andreas Savas Tolia. *AvaTr: One-Shot Speaker Extraction with Transformers*. **Interspeech**, 2021.

Preprints.....

[7] **Md Rifat Arefin**, Ravid Shwartz-Ziv, Ernie Chang, Chinnadhurai Sankar, Rylan Conway, Aristide Baratin, Irina Rish, Adithya Sagar, Patrick Huber. *Delayed Effects of Temporary Weight Decay in Muon Pretraining*. Preprint, 2026 (under review).

[8] Patrick Huber, Ernie Chang, Chinnadhurai Sankar, Rylan Conway, Igor Fedorov, **Md Rifat Arefin**, Adithya Sagar. *Short Data, Long Context: Distilling Positional Knowledge in Transformers*.

Workshop papers.....

[9] **Md Rifat Arefin**, Irina Rish. *When Redundant Probes Help in Strategic Auditing*. EvoRobust Workshop @ **NeurIPS** 2026.

[10] Oscar Skean, **Md Rifat Arefin**, Ravid Shwartz-Ziv. *Does Representation Matter? Exploring Intermediate Layers in Large Language Models*. Machine Learning and Compression Workshop @ **NeurIPS** 2024.

[11] Mohammad-Javad Darvishi-Bayazi, **Md Rifat Arefin**, Jocelyn Faubert, Irina Rish. *VFA: Vision Frequency Analysis of Foundation Models and Human*. Workshop on Foundation Models in the Wild @ **ICML** 2024.

[12] **Md Rifat Arefin**, Vincent Michalski, Pierre-Luc St-Charles, Alfredo Kalaitzis, Sookyoung Kim, Samira E. Kahou, Yoshua Bengio. *Multi-Image Super-Resolution for Remote Sensing Using Deep Recurrent Networks*. EarthVision Workshop @ **CVPR** 2020.

Achievements

- IVADO Ph.D. Excellence Scholarship (2021–2024)
- Notable Reviewer, ICLR 2025
- 1st runner-up, PROBA-V Super-Resolution Competition, European Space Agency
- Travel grant, Deep Learning and Reinforcement Learning Summer School 2019
- IEEE BDS Best Paper Award, International Conference on Computer and Information Technology (ICCIT) 2015
- President's Scholarship during Master's studies
- Government scholarships in Class V, VIII and X

Reviewing and Mentorship

- Reviewer: NeurIPS 2024, 2025, 2026; ICML 2025, 2026; ICLR 2025, 2026, 2027; AISTATS 2024
- Workshop reviewer: ICLR 2024 Workshop on Reliable and Responsible Foundation Models; ICML 2024 Workshop on Foundation Models in the Wild
- Internship mentor for a Mila professional Master's student (May 2024 to December 2024)